CS 306: Media Access Control

Dr. Ayan Mondal

Assistant Professor

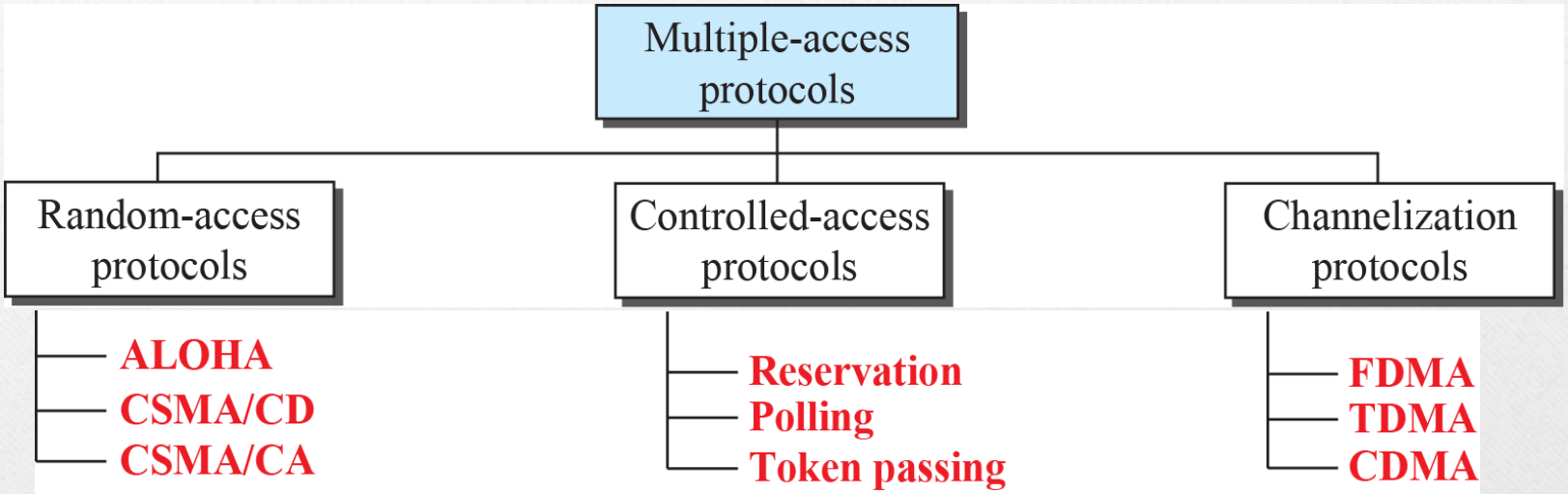
Dept. of CSE, IIT Indore

https://www.iiti.ac.in/people/~ayanm/

Outline

- Random Access

Taxonomy of multiple-access protocols

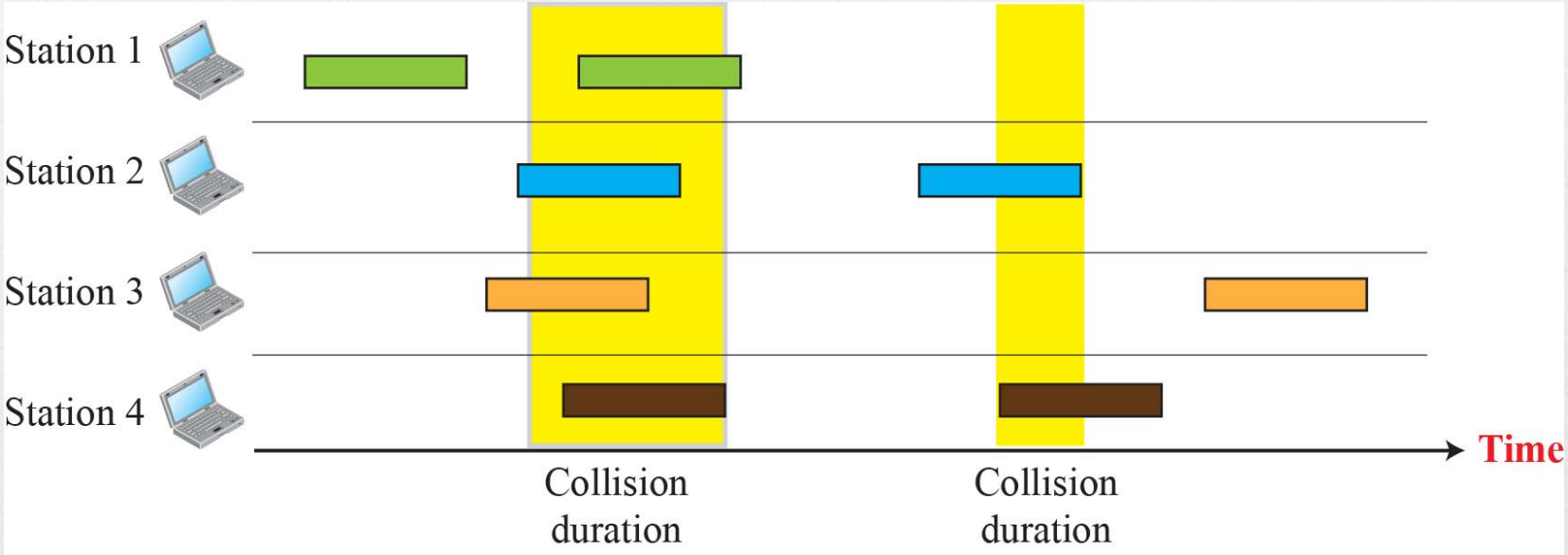


Random Access

- In random-access or contention no station is superior to another station, and none is assigned control over another.
- At each instance, a station that has data to send uses a procedure defined by the protocol to decide on whether to send.
- This decision depends on the state of the medium (idle or busy).

ALOHA

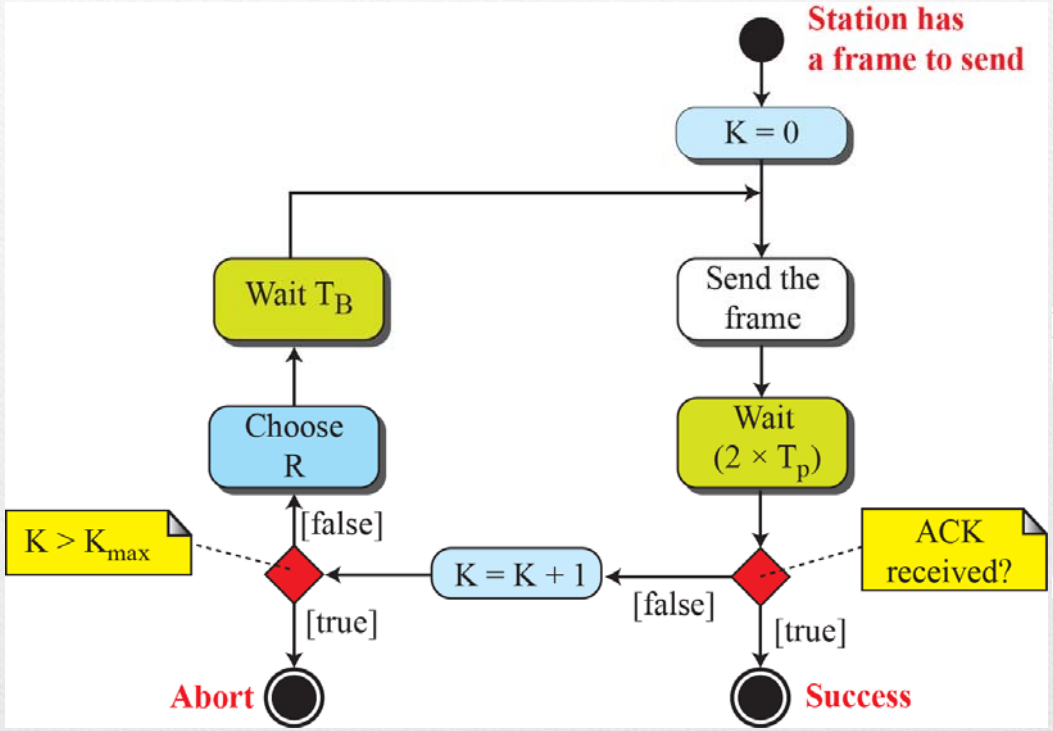
- ALOHA, the earliest random access method, was developed at the University of Hawaii in early 1970.



ALOHA

- It was designed for a radio (wireless) LAN, but it can be used on any shared medium.
- It is obvious that there are potential collisions in this arrangement. The medium is shared between the stations. When a station sends data, another station may attempt to do so at the same time. The data from the two stations collide and become garbled.

Procedure for pure ALOHA protocol



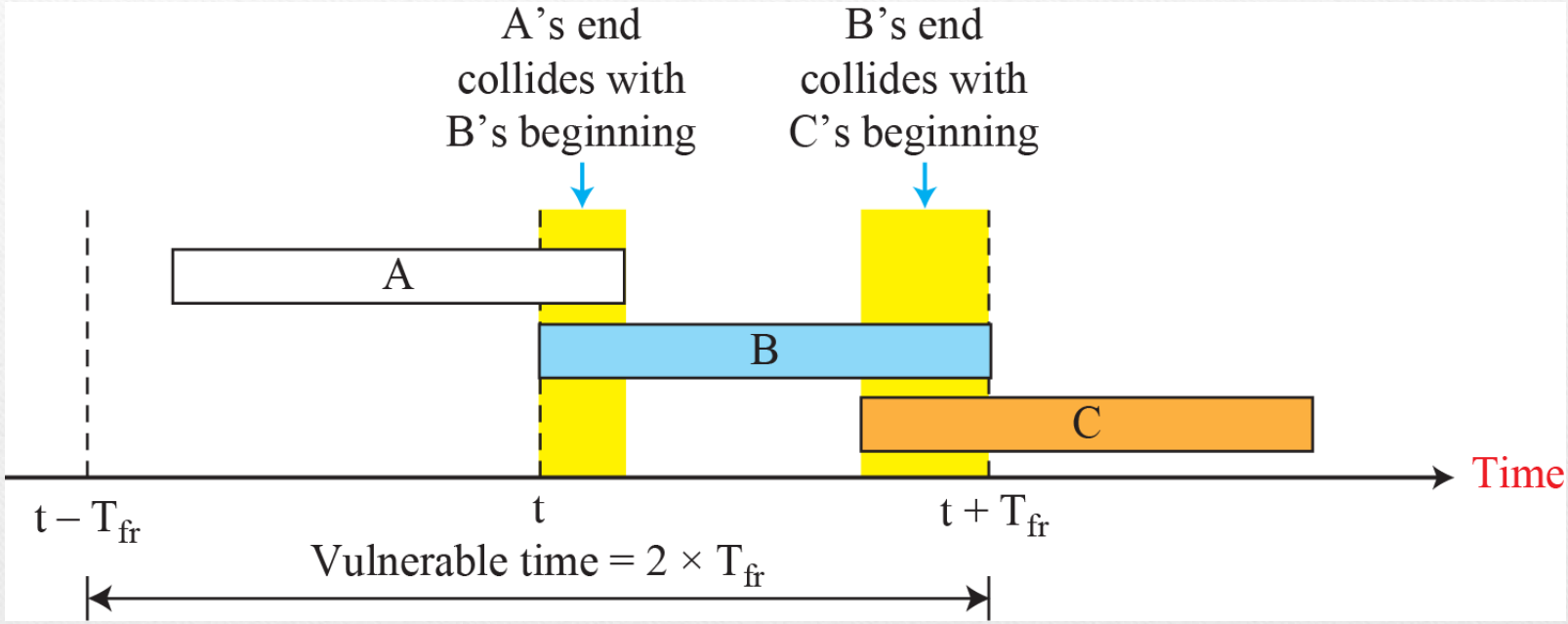
Legend

K : Number of attempts
T_p : Maximum propagation time
T_{fr}: Average transmission time
T_B: (Back-off time): $R \times T_p$ or $R \times T_{fr}$
R : (Random number): 0 to $2^K - 1$

Example

- The stations on a wireless ALOHA network are a maximum of 900 km apart. If we assume that signals propagate at 3×10^8 m/s, we find $T_p = (900 \times 10^3) / (3 \times 10^8) = 3$ ms.
- For $K = 2$, the range of R is $\{0, 1, 2, 3\}$.
- This means that T_B can be 0, 3, 6, or 9 ms, based on the outcome of the random variable R.

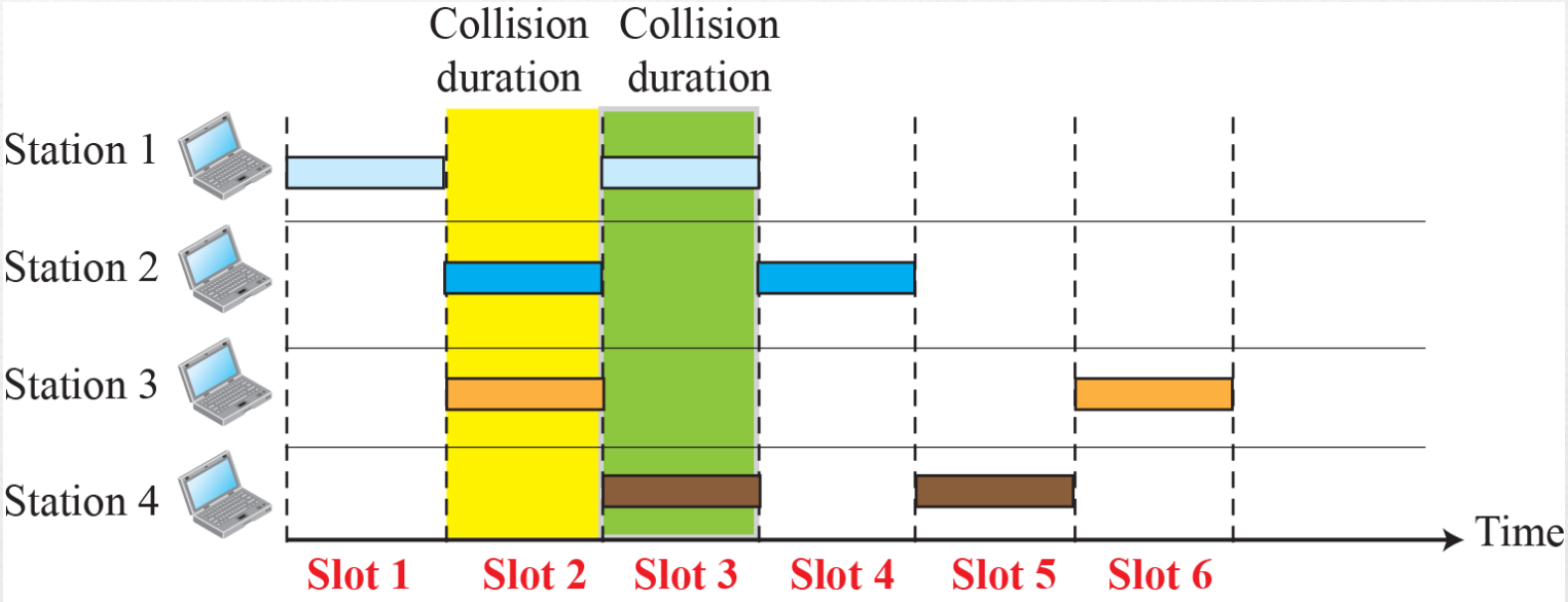
Vulnerable time for pure ALOHA protocol



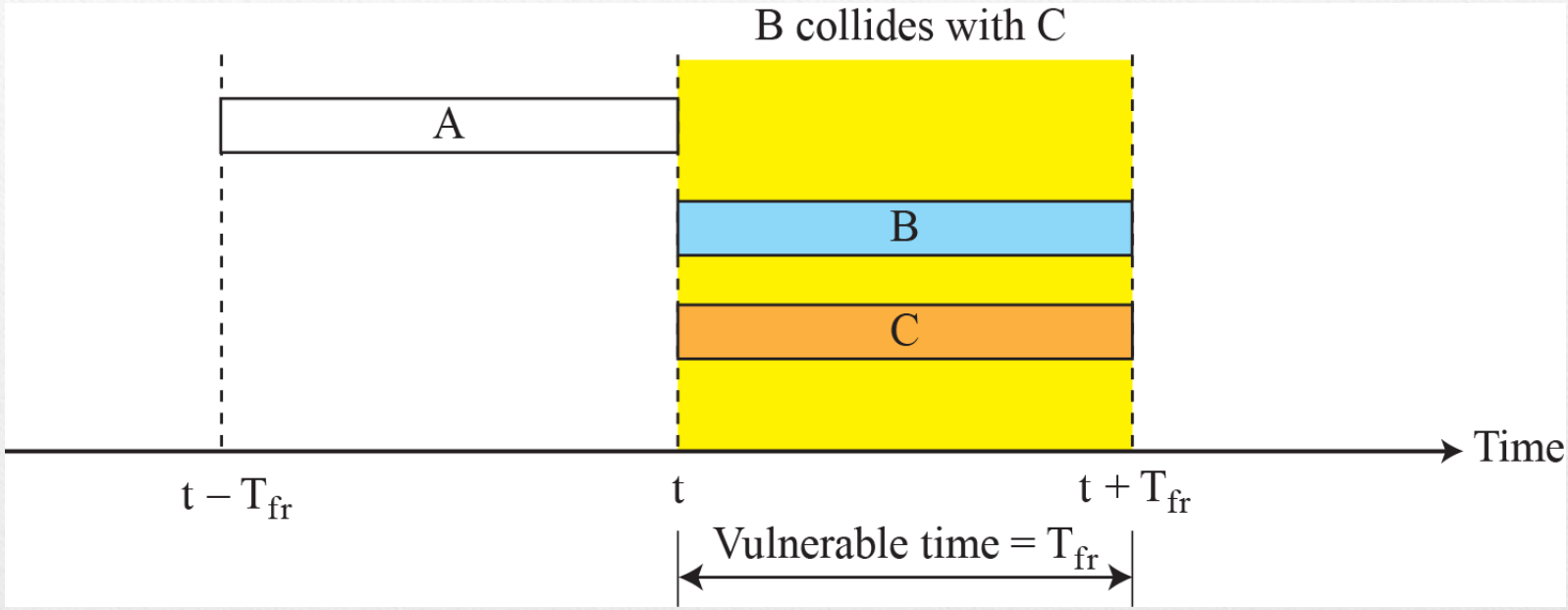
Example

- A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?
- Solution: Average frame transmission time T_{fr} is 200 bits/200 kbps or 1 ms. The vulnerable time is 2×1 ms = 2 ms. This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the period (1 ms) that this station is sending.

Frames in a slotted ALOHA network



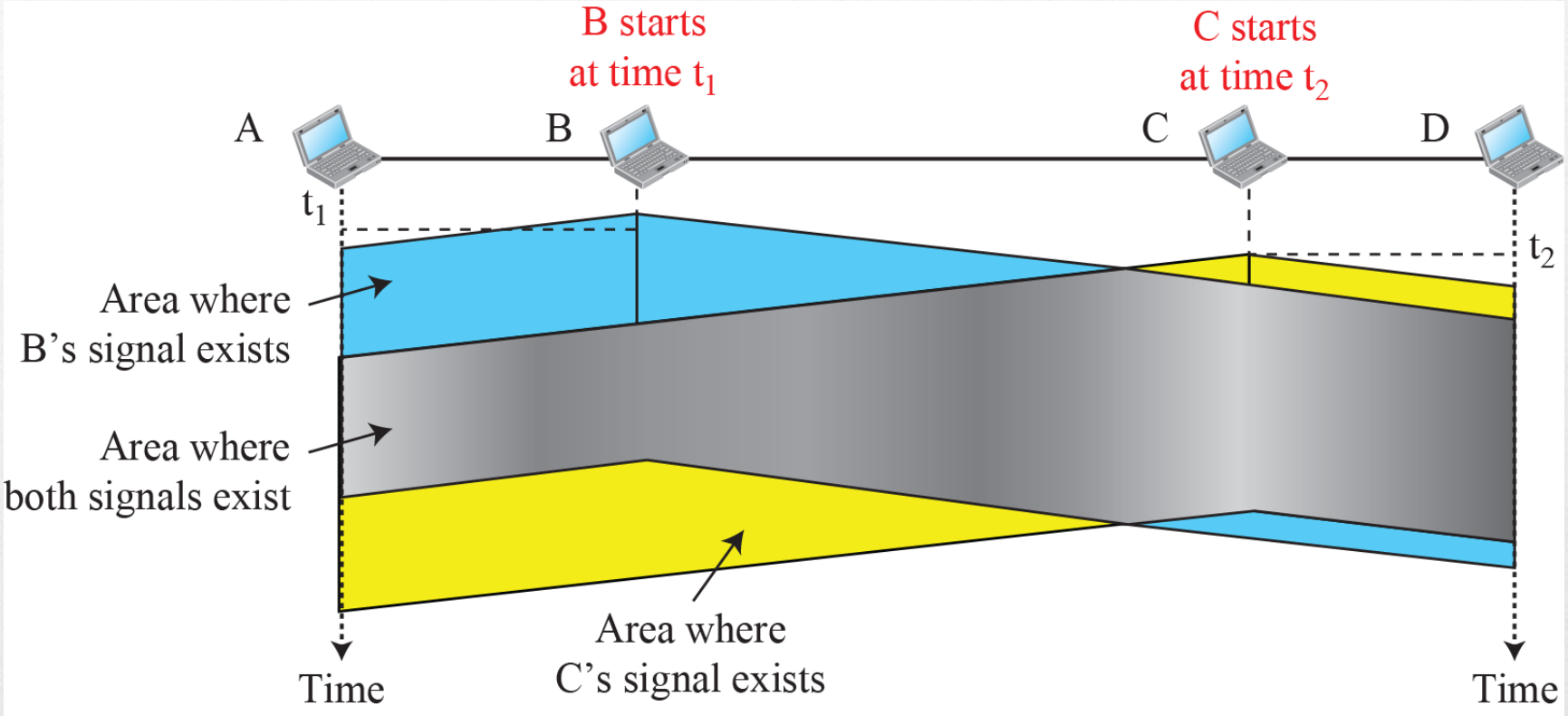
Vulnerable time for slotted ALOHA protocol



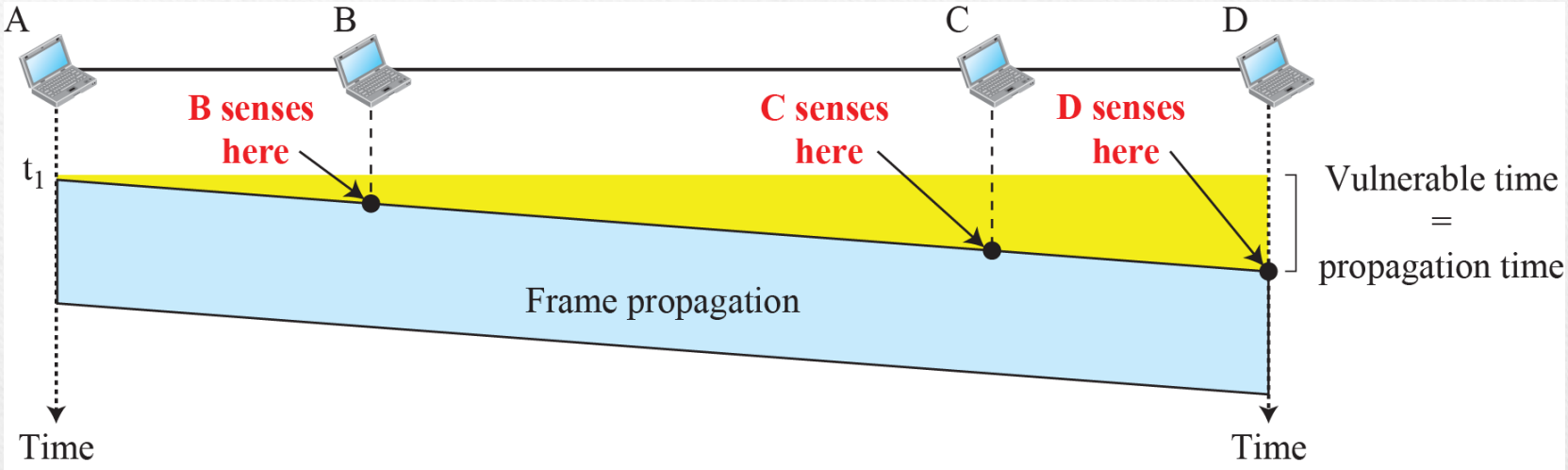
Carrier sense multiple access (CSMA)

- To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed.
- CSMA requires that each station first listen to the medium (or check the state of the medium) before sending. In other words, CSMA is based on the principle "sense before transmit" or "listen before talk."

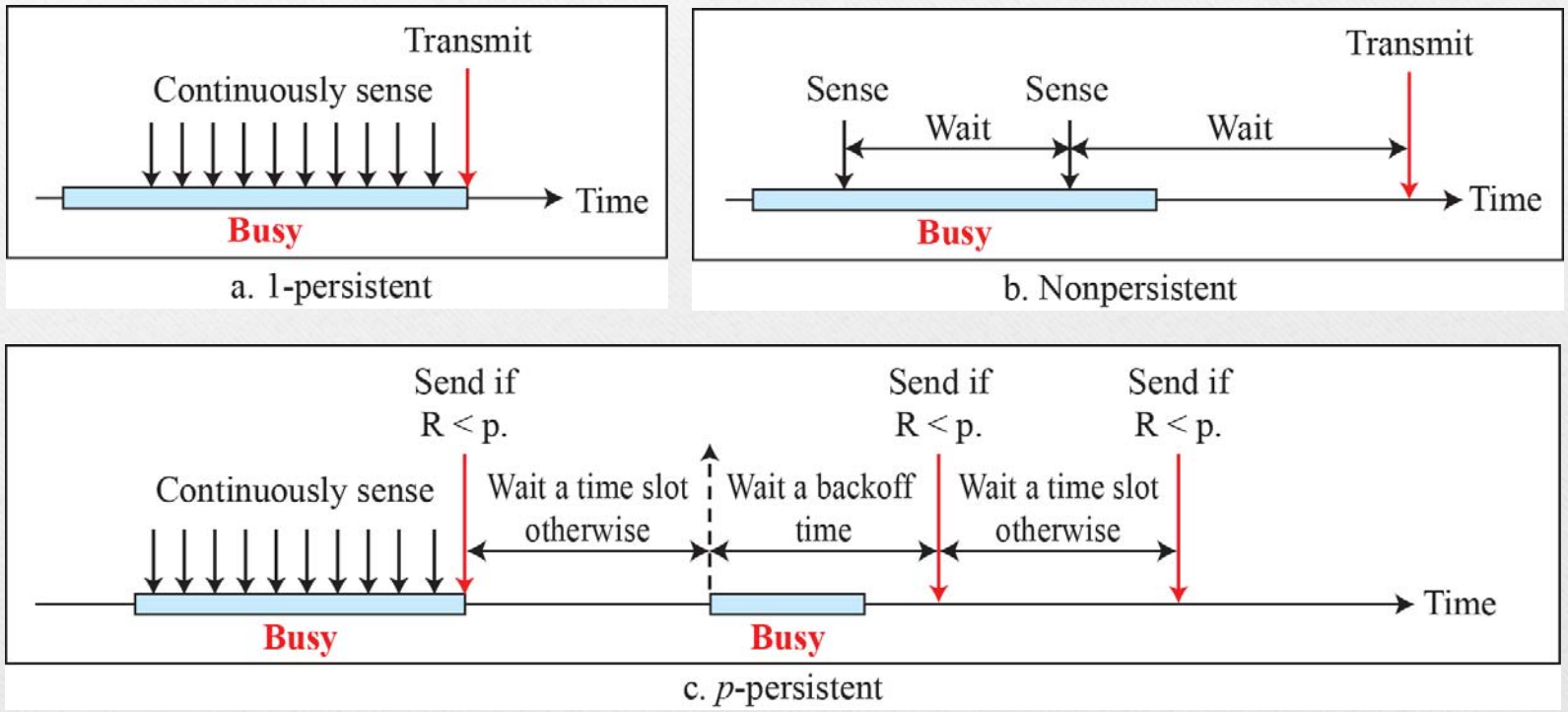
Space/time model of a collision in CSMA



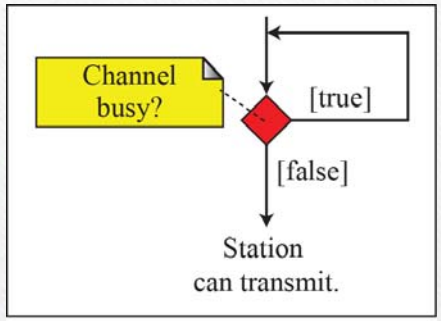
Vulnerable time in CSMA



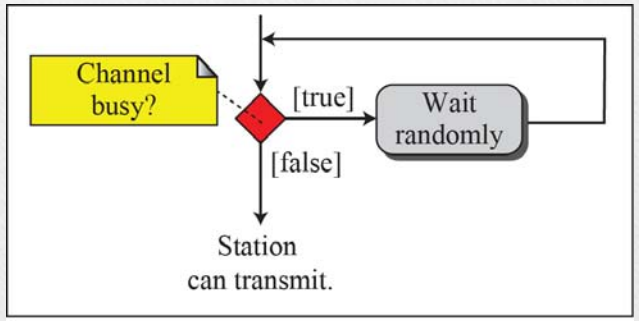
Behavior of three persistence methods



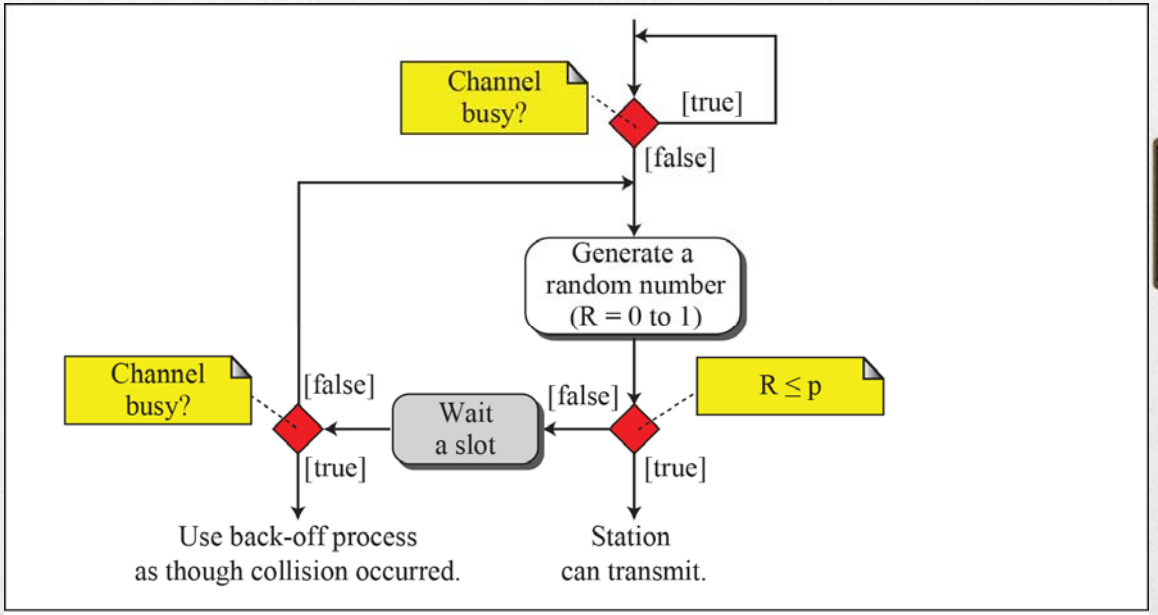
Flow diagram for three persistence methods



a. 1-persistent



b. Nonpersistent

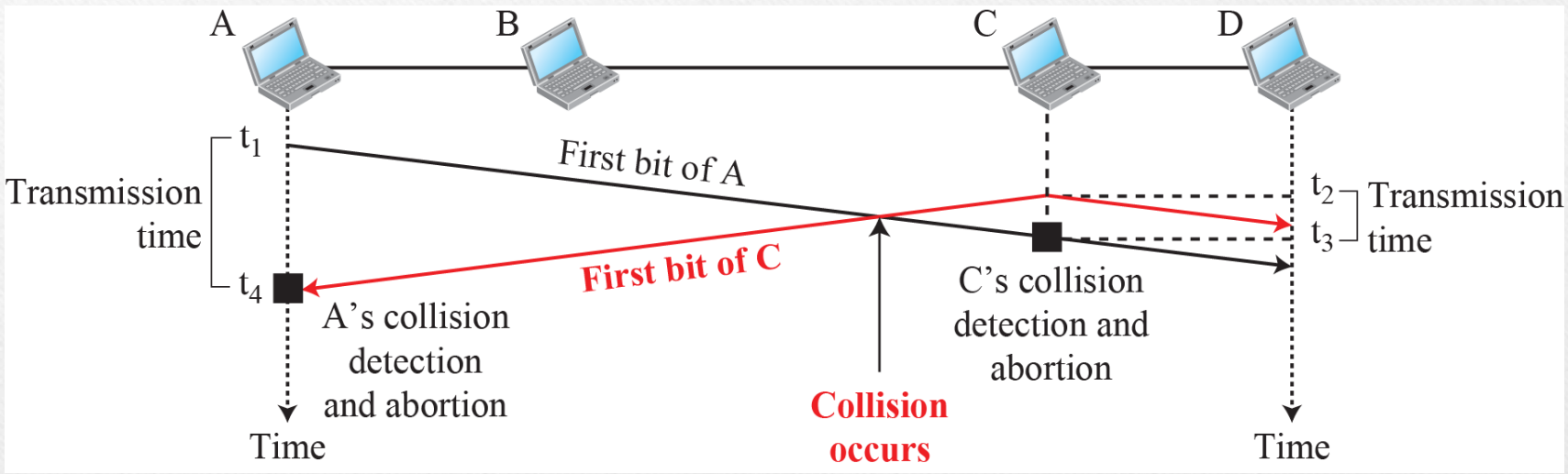


c. *p*-persistent

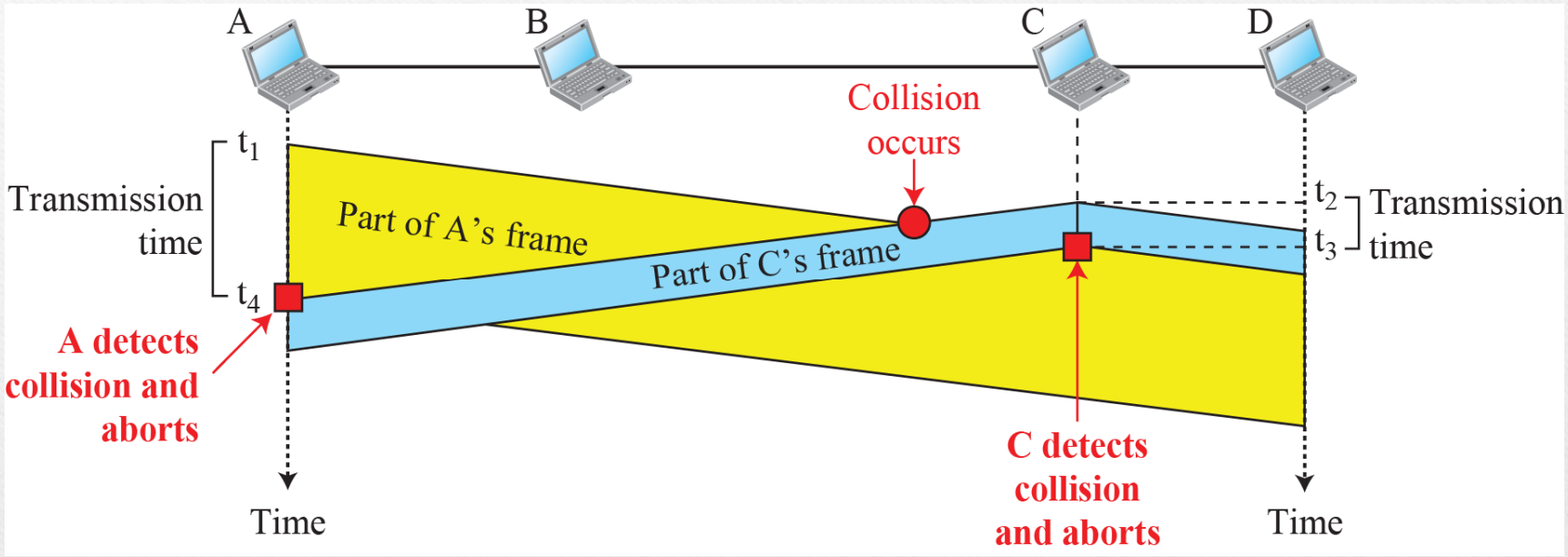
CSMA/CD

- The CSMA method does not specify the procedure following a collision. Carrier sense multiple access with collision detection (CSMA/CD) augments the algorithm to handle the collision.
- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.

Collision of the first bits in CSMA/CD



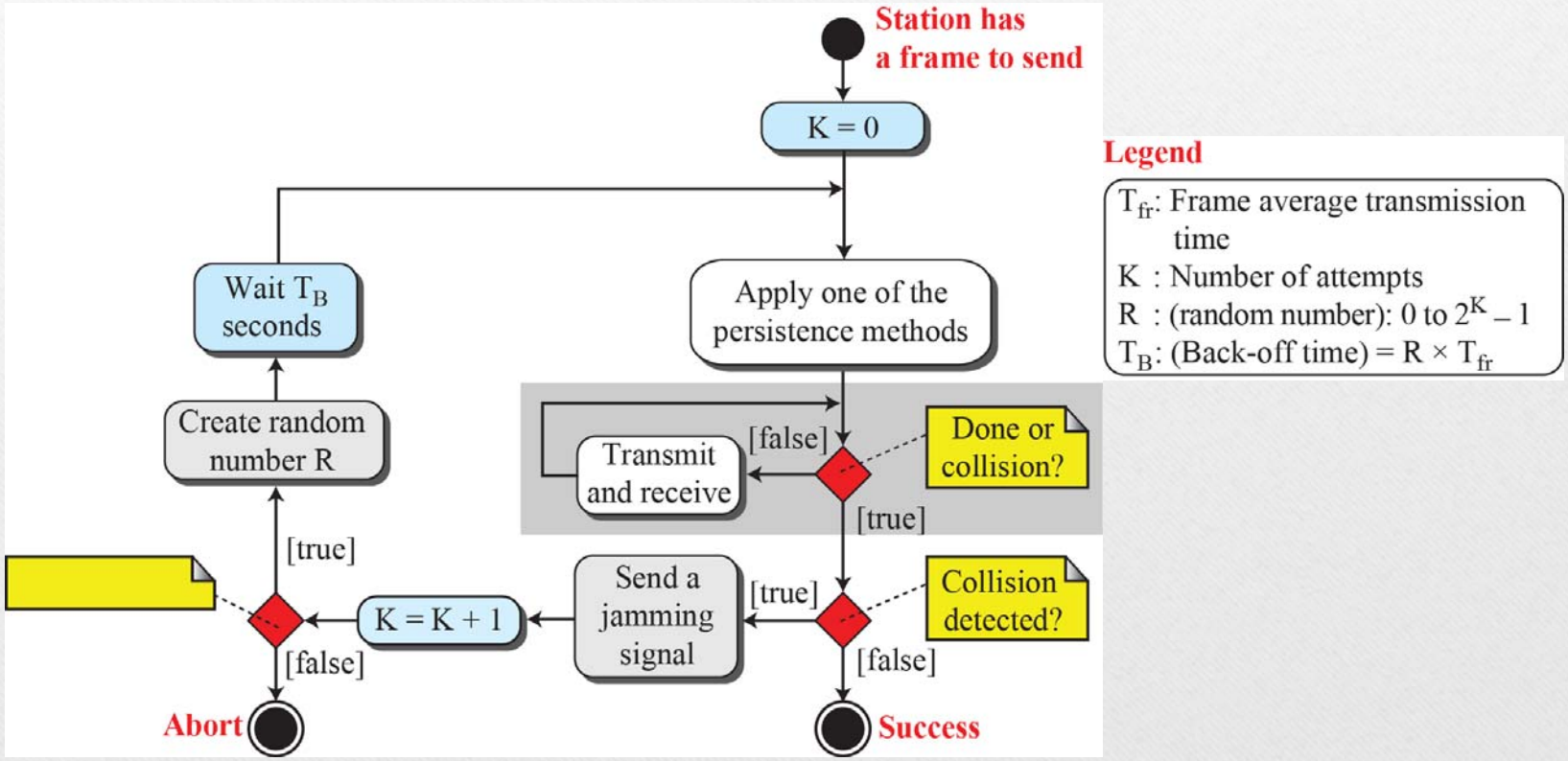
Collision and abortion in CSMA/CD



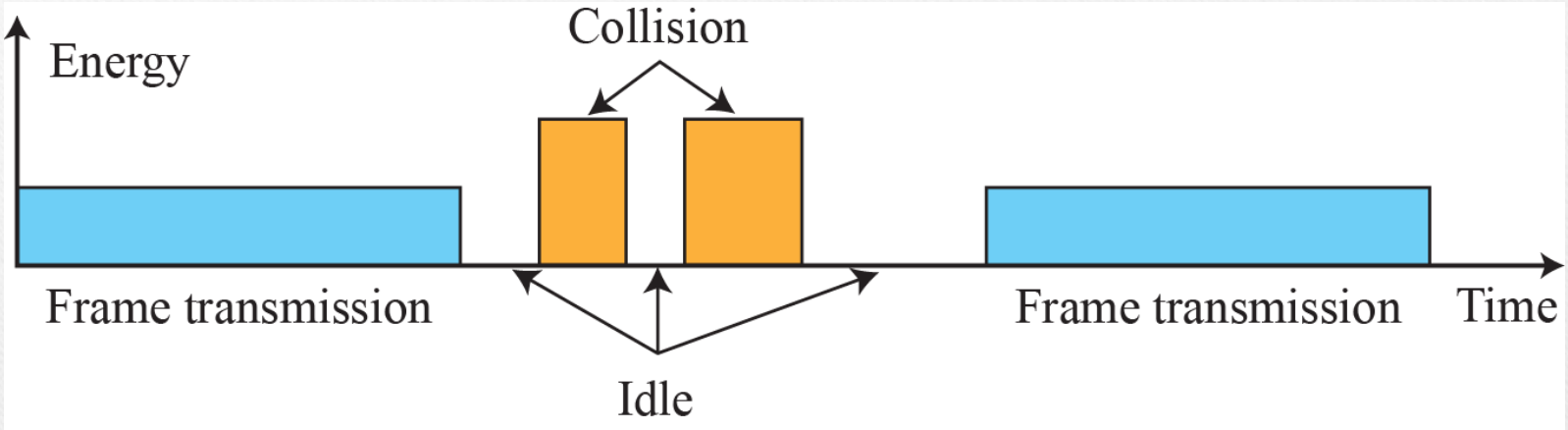
Example

- A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time is 25.6 μs, what is the minimum size of the frame?
- Solution: The minimum frame transmission time is $T_{fr} = 2 \times T_p = 512.2$ μs. This means, in the worst case, a station needs to transmit for a period of 512.2 μs to detect the collision. The minimum size of the frame is 10 Mbps × 512.2 μs = 512 bits or 64 bytes.

Flow diagram for the CSMA/CD



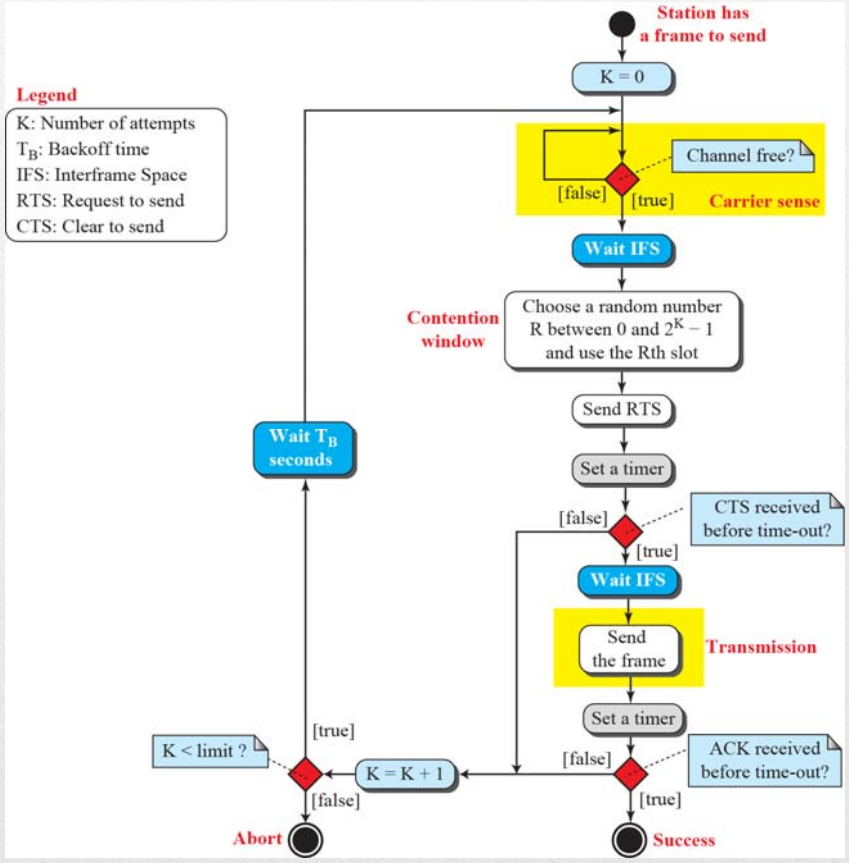
Energy level during transmission, idleness, or collision



CSMA/CA

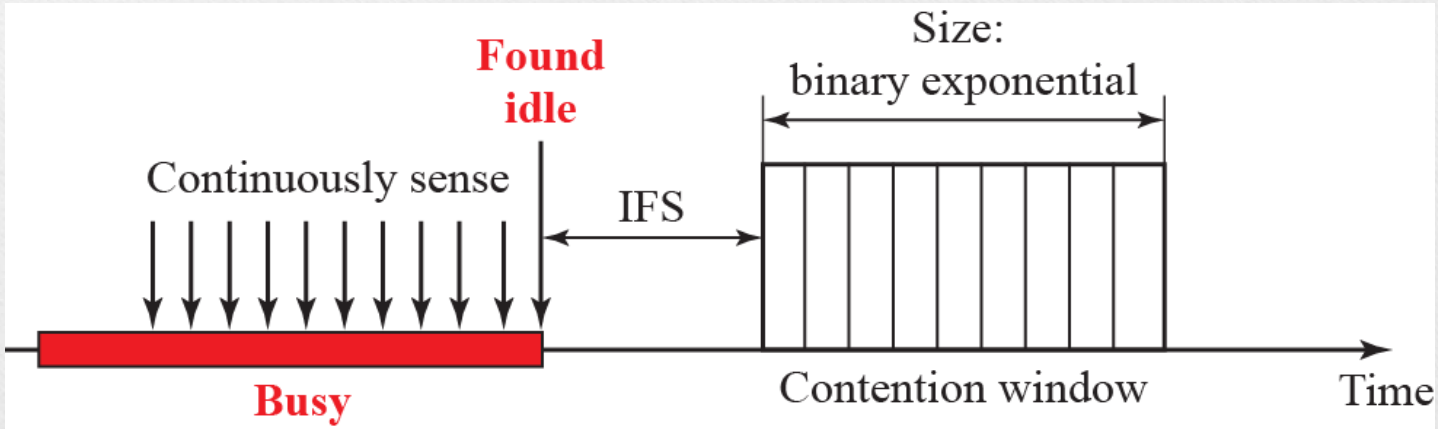
- Carrier sense multiple access with collision avoidance (CSMA/CA) was invented for wireless networks.
- Collisions are avoided using CSMA/CA's three strategies:
  - The interframe space
  - The contention window
  - Acknowledgments

Flow diagram for CSMA/CA

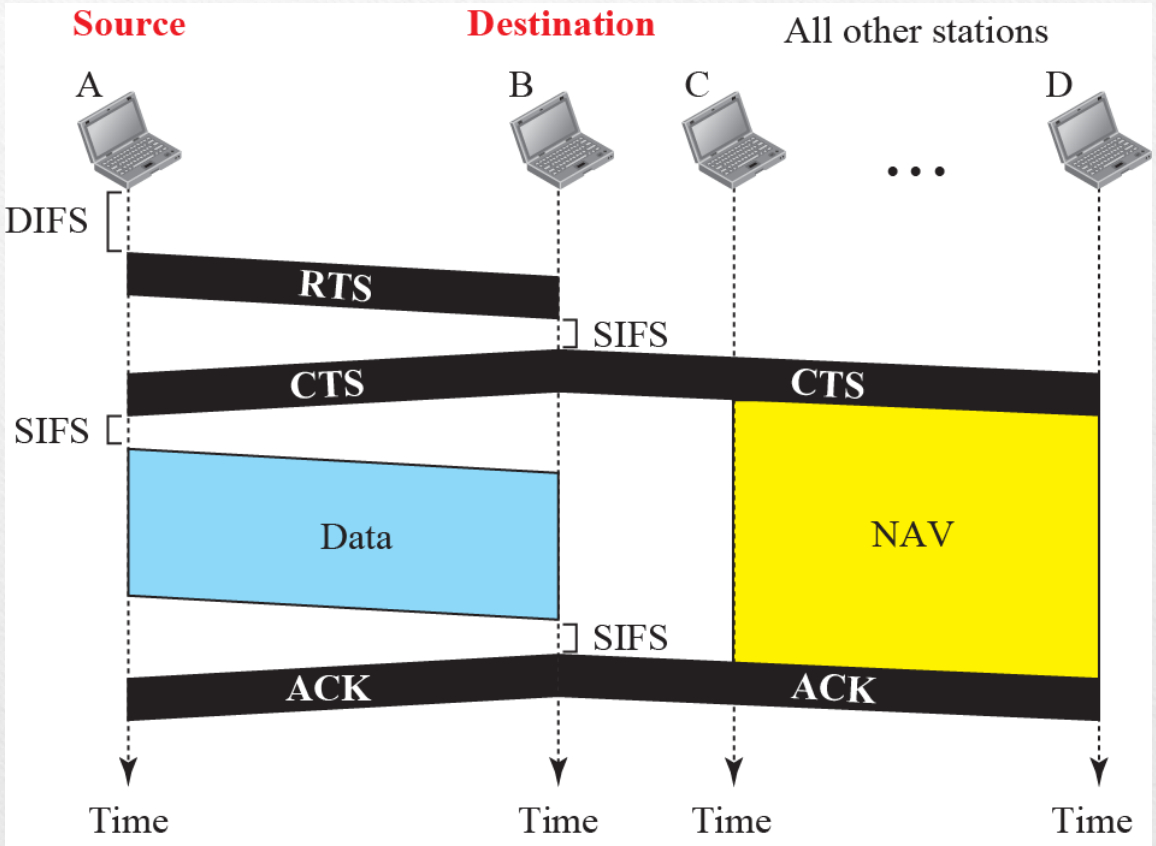


Contention window

- DIFS: DCF Interframe Space
- SIFS: Short Interframe Space



CSMA/CA


Source
Destination
All other stations
A
B
C
D
DIFS
RTS
SIFS
CTS
CTS
SIFS
Data
NAV
SIFS
ACK
ACK
Time
Time
Time
Time

CSMA/CA (Contd.)

- Request to Send (RTS)
- Clear to Send (CTS)
- Network Allocation Vector (NAV)
- Acknowledgement (ACK)

Next Class

- Media Access Control
  - Controlled Access
  - Channelization
- Wired LANs

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ayanm@iiti.ac.in

Room No. 407, POD-1A